Evaluating the Adversarial Robustness of ML Intrusion Detectors in IoT Netw

Research Project Report | Final-Year Cybersecurity Engineering

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1. MOTIVATION & RESEARCH QUESTIONS
Machine Learning Network Intrusion Detection Systems (NIDS) report near-perfect (>99%) accuracy on static benchmarks. In this work, I investigate whether these claims hold against realistic adversarial evasion on the modern CICIoT2023 dataset across 34 attack types, 8 families, and binary tasks.

- RQ1 (Clean Detection): Do Random Forest, XGBoost, and MLP detectors truly generalize, or does class imbalance mask minority-attack detection failure?
- RQ2 (Evasion & Transfer): How much do accuracy and recall degrade under (a) unconstrained PGD, (b) realistic feature-constrained evasion adhering to network physics, and (c) transfer to tree models?
- RQ3 (Defenses & Cost): Can adversarial training and feature squeezing restore robustness, and what clean accuracy penalty is incurred?

2. FORMULATING A REALISTIC THREAT MODEL
Rather than allowing arbitrary mathematical perturbations, the threat model restricts the adversary to physically viable sender-side modifications:
- Controllable (10 features): Packet sizes (Min, Max, AVG, Std, Tot size) can only increase (payload padding). Rates (Rate, Srate) can only decrease. Timings (IAT, flow_duration) can only increase.
- Invariants Enforced: Strict non-negativity, Min <= AVG <= Max, Min <= Tot size <= Max, and Srate <= Rate.
- Fixed (25 features): Network protocols (TCP, UDP, ICMP), application flags, and victim measurements (Drate) are frozen to preserve protocol semantics and connection handshakes.
- Derived (11 features): Tot sum is recomputed as Number * AVG; complex interaction metrics are frozen.

3. SUMMARY OF KEY FINDINGS
- High clean accuracy (>99.0%) is misleading: Multiclass Macro F1 is only 0.70 for trees and 0.59 for MLP due to severe flood volume imbalance overshadowing stealthy attacks.
- Feature-constrained evasion collapses MLP multiclass accuracy from 91.51% to 42.65% (53.40% success rate) simply by padding packet sizes and slowing transmission without altering a single protocol flag.
- Surrogate transfer devastates tree detectors: Unconstrained PGD reduces Random Forest multiclass accuracy from 99.01% to 29.16% and XGBoost from 99.06% to 16.96%.
- Adversarial training restores multiclass robustness to 72.81% under PGD at an 11.11% clean accuracy penalty.
- Feature squeezing proves unsuitable for continuous tabular flow metrics (dropping clean accuracy to 58.95%).

Figure 1: Baseline Clean Accuracy Across Target Granularities (RQ1)

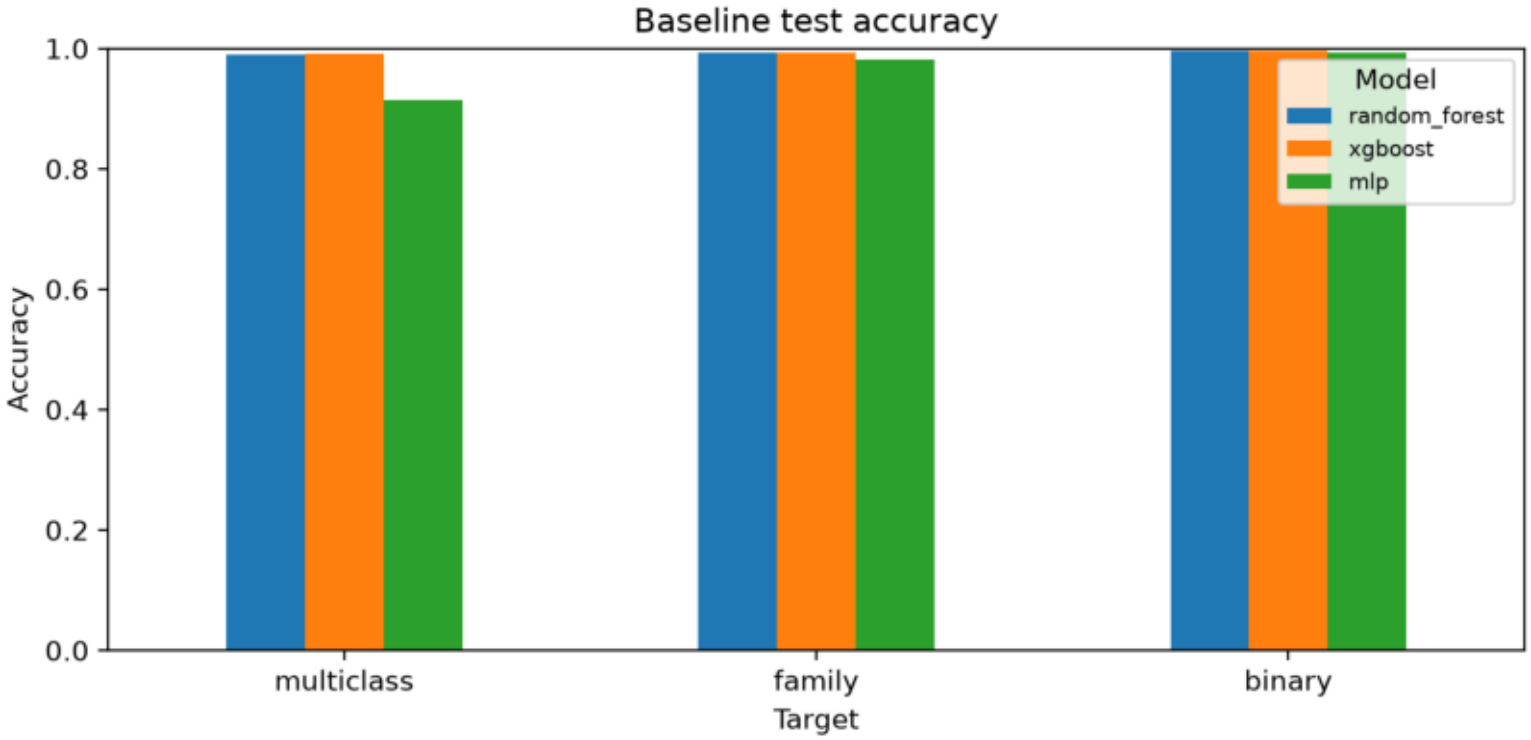


Figure 2: Multiclass Evasion Curves - MLP vs Tree Transfer (RQ2)

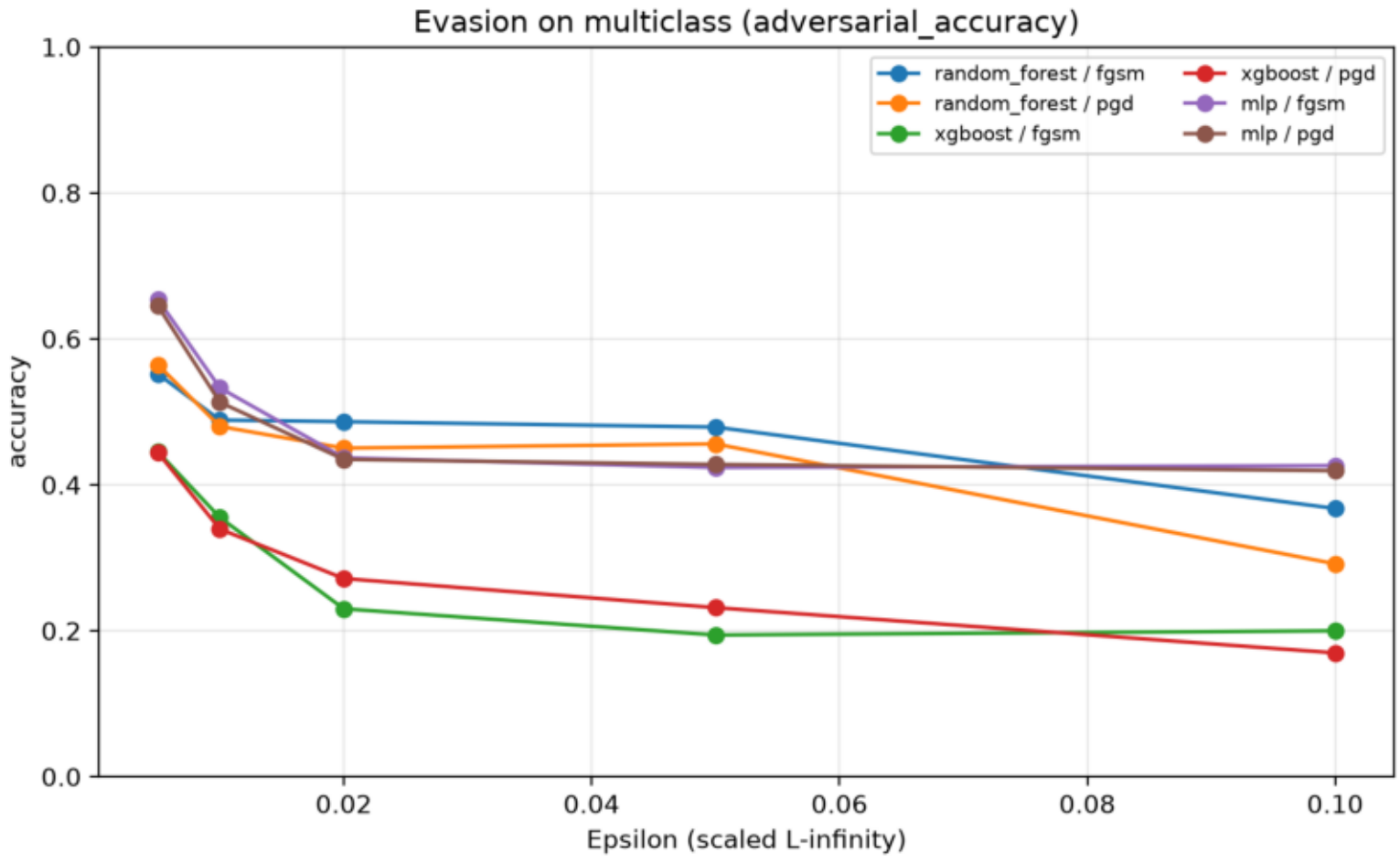


Figure 3: Defense Robustness vs Clean Accuracy Trade-off (RQ3)

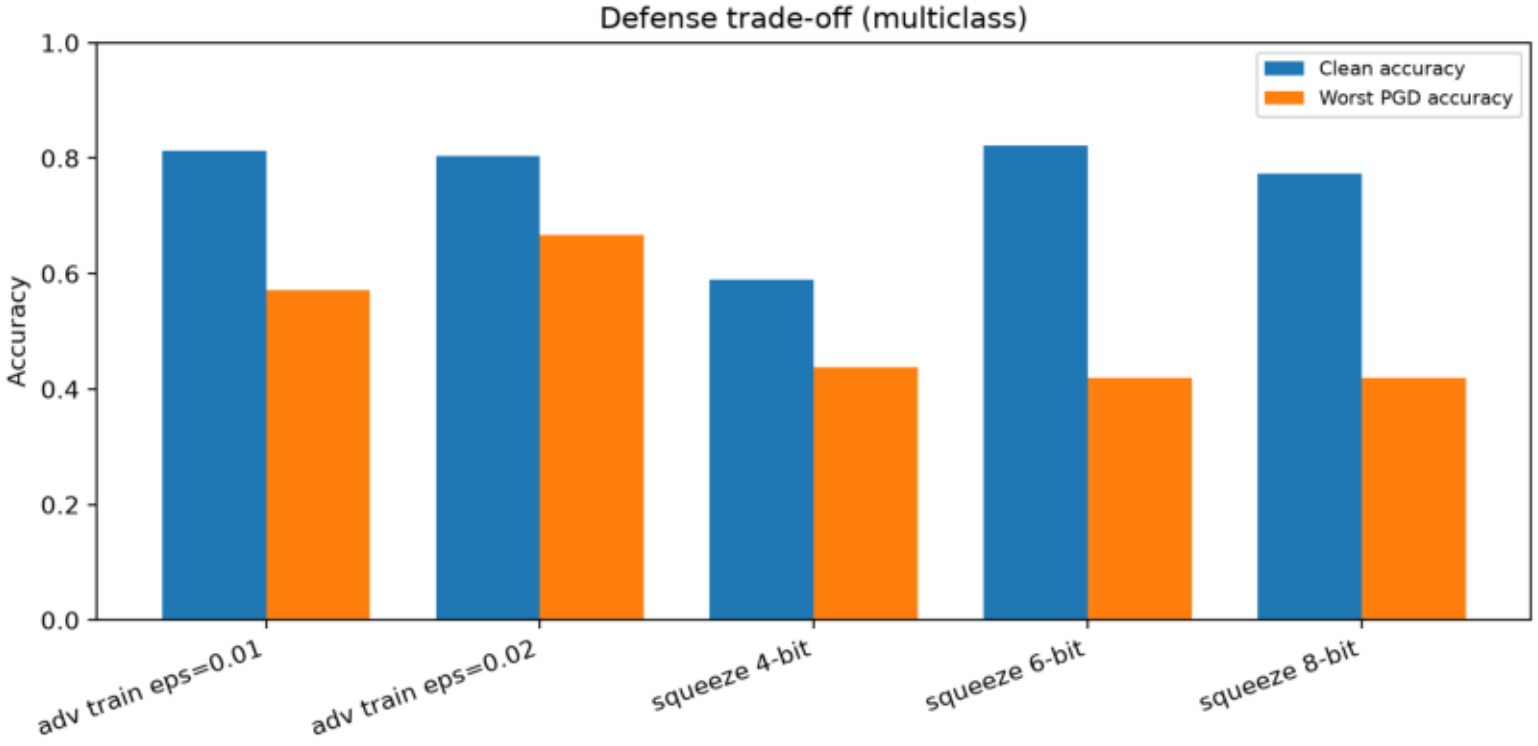


Figure 4: Per-Class Attack Recall Across Baseline Detectors

